Inventory Management System

System Specification Document

PHP + XAMPP · Multi-Department · Role-Based Access

Version 1.0 · Confidential — Internal Use Only

1. System Overview

This document is the complete specification for an inventory management system built with PHP and XAMPP. It covers user roles, department pages, inventory features, the RTS (Return to Sender) module, product costing, POS for walk-in sales, AR credits, and the database design required to support all features.

1.1 Technology Stack

- Backend: PHP (XAMPP / Apache + MySQL)
- Frontend: HTML, CSS, JavaScript (with data tables and Excel export)
- Excel export: PhpSpreadsheet library (via Composer)
- Authentication: PHP sessions with bcrypt password hashing
- Database: MySQL via phpMyAdmin

1.2 Departments

Department	Focus	Notes
Honey	Honey production inventory	Products have size and variant attributes
Paragis	Paragis production inventory	Quantity-based products only
Cacao / BRC	Cacao and BRC production inventory	Quantity-based products only
CSR	Customer Service Representative	Handles RTS only — no production inventory

2. User Roles & Hierarchy

The system has five role types. Each role gets its own homepage and scoped access. Role hierarchy from highest to lowest: Executive > Production Manager > Team Leader > Staff. POS is a separate PIN-based entry with no role in the hierarchy.

2.1 Role Summary

Role	Scope	Key Capabilities
Executive	All departments	Full access. Staff management, system settings, pricing, supplier details, AR overview, all reports, costing module. Cannot be edited by any lower role.
Production Manager	All departments	Cross-department visibility. Views all inventory and reports (except pricing, supplier fields, staff management, system settings). Only role that can log RTS reuse.
Team Leader	Own department	Inputs stock and finished products. Views ALL staff inputs in their dept. Daily stock count. Sets beginning inventory for own dept. Submits edit requests. CSR TL also views RTS records.
Staff	Own department	Inputs stock entries and daily stock count. Deducts stock on shipment. Cannot view other staff entries. CSR staff records RTS parcels only.
POS (PIN)	Walk-in sales only	Accessed via a separate PIN on the login page. Sells products, generates receipts. No access to inventory, reports, or any other module.

2.2 Login Page

The login page (login.php) has two distinct entry points:

- Regular login form — username + password. Routes each role to their own homepage after authentication.
- Walk-in POS button — redirects to pos.php. Requires a separate PIN credential. No access to any system page other than the POS.

These must be two completely separate PHP session types to ensure a POS PIN cannot access inventory pages.

2.3 Role Homepage Pages

Role	Homepage URL	Pages Available
Executive	/exec/dashboard	Dashboard, Staff Management, System Settings, All Dept Reports, Supplier Master, Pricing & AR Overview, RTS Overview, Excel Exports
Prod. Manager	/prodmgr/dashboard	Dashboard (all depts), RTS Overview (all), Log RTS Reuse, All Dept Reports, All Inventory Views, Excel Exports

Role	Homepage URL	Pages Available
Team Leader	/dept/home	Dept Homepage, Input Stock & Finished Products, View All Staff Inputs, Daily Stock Count, Beginning Inventory, Submit Edit Requests. CSR TL also views RTS records.
Staff	/dept/home	Dept Homepage, Input Stock Entries, Daily Stock Count, Submit Edit Requests. CSR Staff: RTS input form only.
POS (PIN)	/pos	Sell Products, Generate Receipt, Buyer Name / Credentials

3. Department Pages

Each department has its own inventory scoped by dept_id. Staff and TL only query rows matching their assigned department. Production Manager and Executives query across all departments.

3.1 Honey, Paragis, Cacao/BRC Departments

These three production departments share the same page structure and features. The only difference is that Honey products have size and variant fields; Paragis and Cacao/BRC are quantity-only.

Feature	Team Leader	Staff
Input incoming stock	Yes — full form	Yes — full form
View other staff inputs	Yes — all dept entries	No — own entries only
Finished product builder	Yes	No
Daily stock count input	Yes	Yes
Deduct stock on shipment	Yes	Yes
Submit edit requests	Yes	Yes
View inventory data table	Yes — all dept rows	No

3.2 Stock Input Fields & Field Visibility

When a stock entry is submitted, fields are shown or hidden based on the logged-in user's role:

Field	Staff	Team Leader	Prod. Manager	Executive
Stock name	Input	Input + View all	View all	Input + View all
Quantity	Input	Input + View all	View all	Input + View all
Batching date	Input	Input + View all	View all	Input + View all
Supplier batch no.	Hidden	Hidden	View	View + Edit
Supplier name	Hidden	Hidden	Hidden	View + Edit
Supplier code	Hidden	Hidden	Hidden	View + Edit
Price	Hidden	Hidden	Hidden	View + Edit

3.3 Finished Products

A finished product is a named product made by combining multiple stocks. Each size/variant of a product has its own formula.

- Executive builds the formula: selects product name, size (if applicable), variant (if applicable), then adds stocks with quantity-per-unit.
- Cost is automatically calculated as the sum of (stock price x qty per unit) using current stock prices at the time of formula creation.
- Honey products require: size (liter, 350ml, 250ml, gallon) and variant (light, amber, dark).
- Paragis and Cacao/BRC products: quantity only, no size/variant fields.
- Staff deduct from the finished product stock when a product is shipped.

3.4 Inventory Reports

Available report periods: daily, weekly, monthly. All report pages include a data table with sort and filter, and an Excel export button.

Report	Who Can Access
Own dept daily / weekly / monthly report	All roles (scoped to their dept)
Cross-department report	Production Manager + Executive only
Report with pricing columns visible	Executive only
Beginning inventory (set monthly)	TL sets it (own dept); Prod. Mgr, Exec view all

3.5 Edit Request & Approval Flow

Staff and Team Leaders cannot directly edit a submitted record. They must submit an edit request:

1. Staff or TL submits an edit request with the original value, proposed value, and reason.
2. Executive receives a notification badge on their dashboard.
3. Executive reviews and approves or rejects.
4. If approved: the record is updated and the change is logged.
5. If rejected: the original record is kept and the requester is notified.

The original record remains locked (read-only) until the request is resolved.

4. CSR Department & RTS Module

CSR (Customer Service Representative) is focused on customer-side operations. For now, their only system function is recording Return to Sender (RTS) parcels.

4.1 What is RTS?

RTS tracks parcels that were shipped but returned. The RTS stock pool is always completely separate from each department's main inventory.

- RTS pool balance = Total recorded minus Total reused (per product, size, variant).
- Honey RTS is tracked by size (liter, 350ml, 250ml, gallon) and variant (light, amber, dark).
- Paragis and Cacao/BRC RTS is tracked by quantity only.

4.2 CSR Staff — Recording RTS

Field	Details
Product type	Dropdown: Honey / Paragis / Cacao/BRC
Size	Dropdown (shown only for Honey): liter, 350ml, 250ml, gallon
Variant	Dropdown (shown only for Honey): light, amber, dark
Quantity returned	Numeric input
Date recorded	Date picker (defaults to today)
Parcel details / notes	Text area (optional)

4.3 RTS Visibility

Role	RTS Access
CSR Staff	Input only — cannot view others' entries
CSR Team Leader	View all CSR RTS entries + running totals per product
Production Manager	View all RTS across all products + can log reuse
Executive	View all RTS across all products (read-only overview)
All other roles	No access to RTS at all

4.4 Production Manager — Logging RTS Reuse

The Production Manager is the only role that can mark RTS stock as reused. Reuse form fields:

- Product type (Honey / Paragis / Cacao/BRC)
- Size (if Honey)
- Variant (if Honey)
- Quantity being reused

- Date of reuse

The form shows the current RTS pool balance before the Prod. Manager inputs the quantity. Validation prevents reuse qty from exceeding the available balance. Every reuse action is logged in rts_reuse_log with timestamp and account.

5. Product Costing Module

This module is Executive-only. It allows the executive to set stock prices, build product formulas, and track how production costs change over time with green/red visual comparisons.

5.1 Stock Price Management

- Executives set and update the price of each stock item at any time.
- Every price change is saved to stock_price_history: old price, new price, changed by (account), changed at (timestamp).
- The stock list shows the current price and a comparison badge vs. the previous price (green if lower, red if higher).
- A "History" link on each stock opens its full price change log.

5.2 Product Formula Builder

For each product (and for each size/variant combination for Honey), the executive builds a formula:

6. Select the product name, size (if Honey), and variant (if Honey).
7. Add stocks to the formula, with a quantity-per-unit field for each stock.
8. The system calculates: Production Cost = sum of (stock current price x qty per unit).
9. This calculated cost and each stock's price at that moment are saved as the baseline.

5.3 Automatic Cost Recalculation

Whenever a stock price is updated, the system automatically:

10. Finds all product formulas that use that stock.
11. Recalculates each product's total production cost using the new stock price.
12. Compares the new cost to the previous cost history entry.
13. Writes a new row to product_cost_history with old cost, new cost, the triggering stock, and timestamp.
14. The product page immediately shows the updated cost with a green (lower) or red (higher) badge vs. the baseline.

5.4 Product Cost View (Executive)

Column	Description
Stock used	Name of each stock in the formula
Qty per unit	How much of that stock goes into one unit of this product
Baseline price	Stock price when the formula was first created (frozen reference)
Current price	Stock's live price today
Line cost (current)	Current price x qty per unit
vs. baseline	Green badge if cheaper than baseline, red badge if more expensive

At the bottom: total production cost per unit (current) vs. total baseline cost, with an overall green/red comparison badge.

A "View cost history" button opens the full cost change log for that product.

5.5 Suggested Addition: Selling Price & Margin

Consider adding a selling price field on each product (Executive-set). The product cost page can then show: Margin = Selling Price minus Production Cost. This immediately shows how much a cost increase eats into profit.

6. POS — Walk-in Sales

The POS is a separate page (pos.php) accessible via a PIN entered on the login page. It is designed for tablet use with large touch targets and simple navigation.

6.1 POS Features

- Select product, size, variant, and quantity.
- Price auto-fills from the product's set selling price.
- Input buyer name or credentials.
- Generate and print/save receipt.
- Each sale deducts from the finished product's inventory.

6.2 POS Constraints

- No access to any inventory, report, RTS, or costing page.
- PIN session is completely separate from regular role sessions.
- POS sales data is visible to Prod. Manager and Executives in their report pages.
- Excel export of POS sales data available for Prod. Manager and Executive.

7. AR Credits (Accounts Receivable)

When any staff member, TL, or Executive takes a product from the system for personal use, the system automatically records it as a credit (debt) against that person's account.

7.1 How Credits Work

- Taking a product logs a credit entry: who took it, what product, quantity, price at time of taking, date.
- The credit remains outstanding until the person pays it back.
- The Executive marks a credit as paid ("uncredited"), clearing it from the outstanding balance.
- The full credit history (paid and unpaid) is always saved and viewable.

7.2 Credit Visibility

Role	AR Access
Staff / TL	Can see their own credits only
Production Manager	Can see credits across all departments
Executive	Full AR overview across all staff and departments + mark as paid

- Excel export available on all AR credit pages.
- Data table with filter by person, date, status (outstanding / paid).

8. Database Schema

All tables use MySQL. Build in this order: users and roles first, then departments, then inventory tables, then RTS, then costing, then POS, then AR.

8.1 Users & Roles

Table	Key Fields
roles	id, name (executive / prod_manager / tl / staff / pos), created_at
departments	id, name (honey / paragis / cacao_brc / csr), created_at
users	id, name, email, password_hash, role_id (FK), dept_id (FK), is_active, created_at, updated_at
pos_pins	id, pin_hash, label, is_active, created_at

8.2 Inventory

Table	Key Fields
stocks	id, name, dept_id (FK), unit, current_price, supplier_name, supplier_code, created_by (FK users), created_at
stock_entries	id, stock_id (FK), dept_id (FK), qty, batching_date, supplier_batch_no, entered_by (FK users), created_at
stock_deductions	id, stock_id (FK), dept_id (FK), qty, reason, deducted_by (FK users), created_at
daily_stock_counts	id, stock_id (FK), dept_id (FK), qty_remaining, counted_by (FK users), count_date
beginning_inventory	id, dept_id (FK), stock_id (FK), qty, set_by (FK users), month_year, created_at
edit_requests	id, table_name, record_id, field_name, old_value, new_value, reason, requested_by (FK users), status (pending/approved/rejected), reviewed_by (FK users), reviewed_at, created_at

8.3 Finished Products & Costing

Table	Key Fields
products	id, name, dept_id (FK), size (nullable), variant (nullable), selling_price, baseline_cost, created_by (FK users), created_at
product_formula	id, product_id (FK), stock_id (FK), qty_per_unit, price_at_creation, created_at
stock_price_history	id, stock_id (FK), old_price, new_price, changed_by (FK users), changed_at

Table	Key Fields
product_cost_history	id, product_id (FK), old_cost, new_cost, triggered_by_stock_id (FK), recalculated_at

8.4 RTS Module

Table	Key Fields
rts_records	id, product_type (honey/paragis/cacao_brc), size (nullable), variant (nullable), qty, notes, recorded_by (FK users), record_date, created_at
rts_reuse_log	id, product_type, size (nullable), variant (nullable), qty_reused, reused_by (FK users), reuse_date, created_at

RTS pool balance: SELECT product_type, size, variant, SUM(qty) FROM rts_records GROUP BY product_type, size, variant — minus the same grouping from rts_reuse_log.

8.5 POS & AR Credits

Table	Key Fields
pos_sales	id, product_id (FK), qty, unit_price, total, buyer_name, buyer_credentials, sold_by_pin (FK pos_pins), sale_date, created_at
pos_receipt_items	id, sale_id (FK), product_id (FK), qty, unit_price, line_total
ar_credits	id, user_id (FK), product_id (FK), qty, price_at_time, total_amount, status (outstanding/paid), taken_at, paid_at, marked_paid_by (FK users), created_at

9. Recommended Build Order

Build each phase completely before starting the next. Do not start Phase 2 without Phase 1's middleware working.

Phase	What to Build	Why First
Phase 1	Users, roles, departments, login page, session middleware, role-based routing, permission gate function	Everything else depends on knowing who is logged in and what they can access
Phase 2	Stock input, stock entries table, field visibility by role, data tables, edit request + approval flow	Core inventory input before anything reads from it
Phase 3	Finished product builder, product formula, stock deductions on shipment, beginning inventory	Needs stock data from Phase 2
Phase 4	Stock price history, product costing module, auto-recalculation on price update, cost history log	Needs products from Phase 3
Phase 5	Daily stock count, inventory reports (daily/weekly/monthly), Excel exports via PhpSpreadsheet	Needs a full inventory to report on
Phase 6	CSR RTS input, RTS reuse by Prod. Manager, RTS visibility rules, RTS running balance	Standalone module, can be built any time after Phase 1
Phase 7	POS page (PIN login, product selection, receipt, sale deduction)	Needs finished products from Phase 3
Phase 8	AR credits (auto-log on product take, mark as paid, visibility by role, Excel export)	Add last once all product and user flows are stable

Key implementation reminders:

- Use a single requireRole(["executive"]) middleware function on every route — never check roles inline per page.
- Every inventory query for Staff and TL must include WHERE dept_id = $_SESSION['dept_id'].
- Use PHP transactions when deducting stock and logging the deduction so they can never get out of sync.
- For the RTS reuse form, run a live AJAX balance check before allowing submission.
- The cost recalculation loop runs inside the same PHP transaction as the stock price update.
- For the POS, use a separate session key (e.g. $_SESSION['pos_authenticated']) so it cannot bleed into staff sessions.

End of Specification — Version 1.0